Data Driven Analysis of University Past Papers & Lecture Content for Curriculum Optimization

25-26J-089

Introduction to the Problem

Misalignment between curriculum, lecture content, and exams.

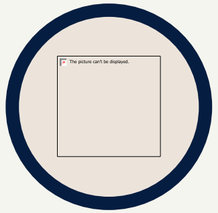
Students focus on less relevant topics, missing key tested areas.

Leads to inefficient preparation and poor outcomes.

Manual analysis of past papers is subjective, inconsistent, and slow.

Need for a data-driven, automated solution.

Social or research impact of the solution

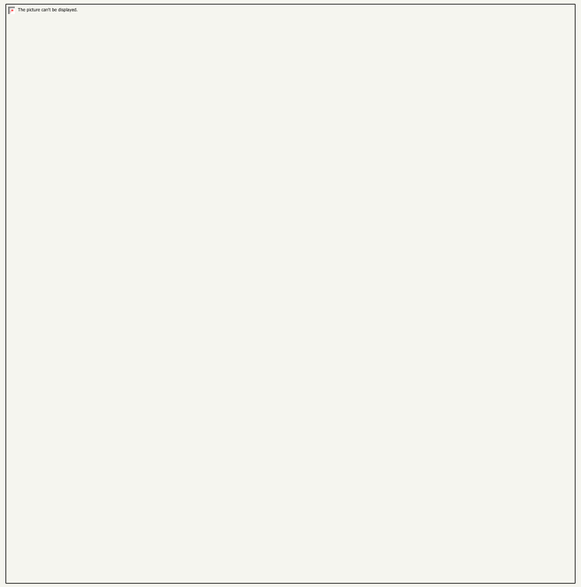


Social Impact

- Improved Educational Equity
- Enhanced Student Well-being and Efficiency

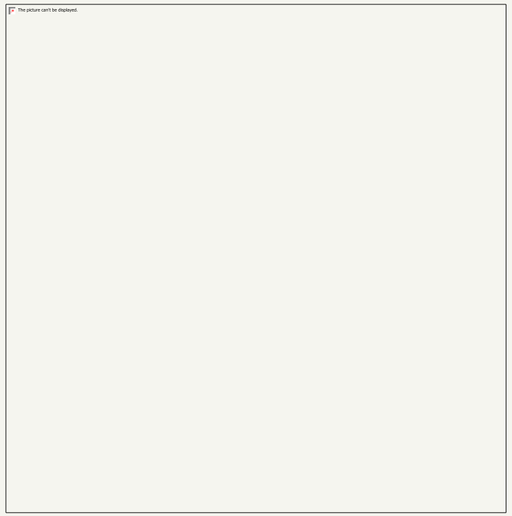
Research Impact

- Future Scalability and Cross-Disciplinary Application
- Contribution to Personalized Learning Methodologies



Previous research & Proposed Solutions

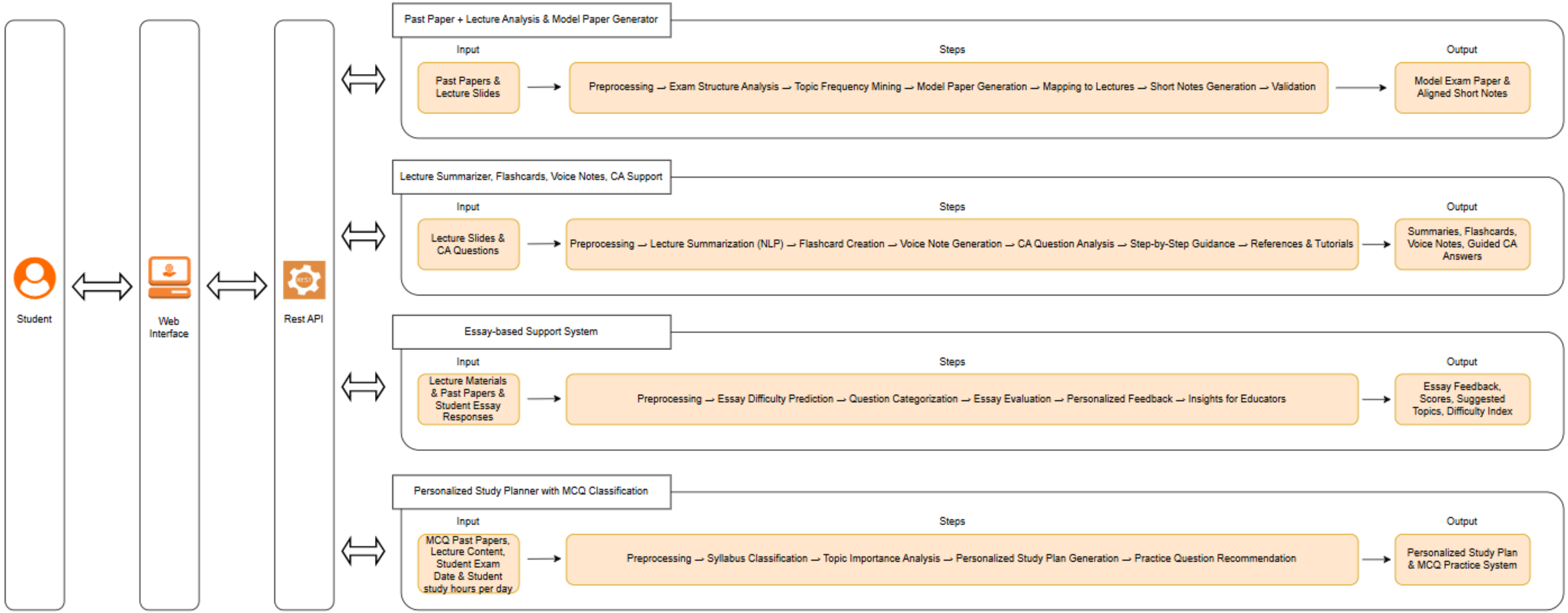
Focus Area	Research Paper 1 (Du et al., 20
Main Objective	Curriculum optimization in highe education
Key Contribution	Improves decision-making for curriculum planning



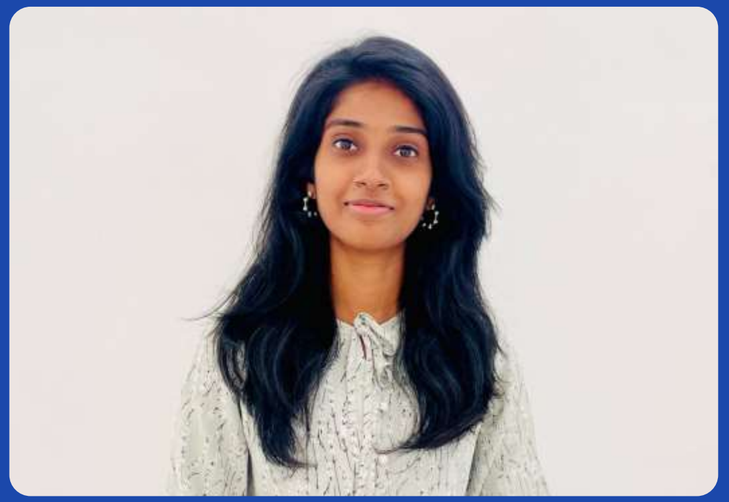
Constraints and Limitations

- Data Dependency
- Model Complexity & Resource Constraints
- Generalizability & Domain Adaptation

Overall System Diagram



AI-Supported Development of Model Exam Paper and Aligned Short Notes



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PROVEN GAP & CREATIVE SOLUTION

Proven Gap

focused only

- Automatic question generation (focused only on single questions)
- Ignoring exam-structure fidelity (sections, marks, question types)
- Generic summaries & resources
- No prioritization of high-frequency topics

Creative

- Authentic Model Exam Paper Generation
- Comprehensive Exam Analysis (uncover recurring topics, question types, and mark distributions.)
- Concise, Syllabus-Aligned Short Notes
- High-Frequency Topic Prioritization
  - most frequently tested areas for efficient revision

KNOWLEDGE GAP & SHORTCOMINGS OF EXISTING SYSTEMS

Knowledge Gap

- No research/system has combined exam-structure analysis + short notes generation.
- Lack of syllabus-aligned notes linked to frequently tested topics.
- Absence of authentic model exam papers

Feature	Research 5 (AQG)	Research 6 (Question Generation + Answer Grading)	Research 7 (Summarization Systems)
Automatic Question Generation	✅	✅	❌
Exam Structure Fidelity (sections, marks, formats)	❌	❌	❌
Summarization (concise notes)	❌	❌	✅ (generic only)

DOMAINS OF KNOWLEDGE

Data Science & Analytics

Perform exam trend and frequency analysis

Machine Learning & Transformers

Use Sentence-BERT / DistilBERT/ FLAN-T5 models for semantic mapping, summarization, and question alignment.

Natural Language Processing

Apply keyword extraction, clustering, and text summarization to generate syllabus-aligned short notes.

Educational Technology & Learning Analytics

Ensure curriculum alignment with exam structures to support focused and efficient student revision.

TOOLS & TECHNOLOGY

Similarity search
FAISS

Programming Languages
Python
Javascript

Database
Mongo DB

Text Processing Libraries
NLTK, spaCy, Regex

Frameworks
Streamlit
Flask

Project Management
Jira

Other tools
Git
Draw.io

Data handling tools
Numpy
Pandas

NLP & ML Models
DistilBERT, FLAN-T5, SBERT

FRAMEWORK & VALIDATION

Framework (Technologies per Step)

Preprocessing
- Cleaning & Tokenization: NLTK / spaCy
- Normalization: regex, lemmatization

Exam Structure Analysis
- Regex
- spaCy (if regex fails)

Topic Frequency & Trend Mining
- TF-IDF (scikit-learn) → keyword frequency
- YAKE / KeyBERT → keyphrase extraction

Model Exam Paper Generation
- FAISS (With Sentence-BERT embeddings) → avoid duplicates
- FLAN-T5 (LoRA fine-tuned) → paraphrasing for novelty
- DistilBERT (fine-tuned) → difficulty classification

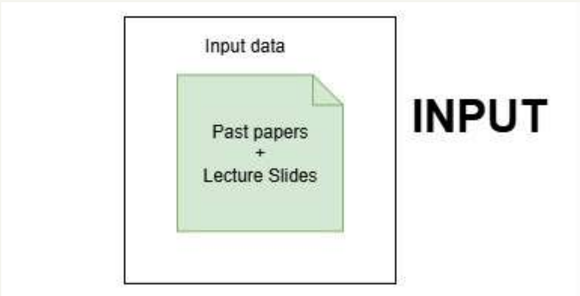
Semantic Mapping to Lecture Content
- Sentence-BERT embeddings → convert text to vectors
- FAISS → similarity search
- Thresholding with cosine similarity

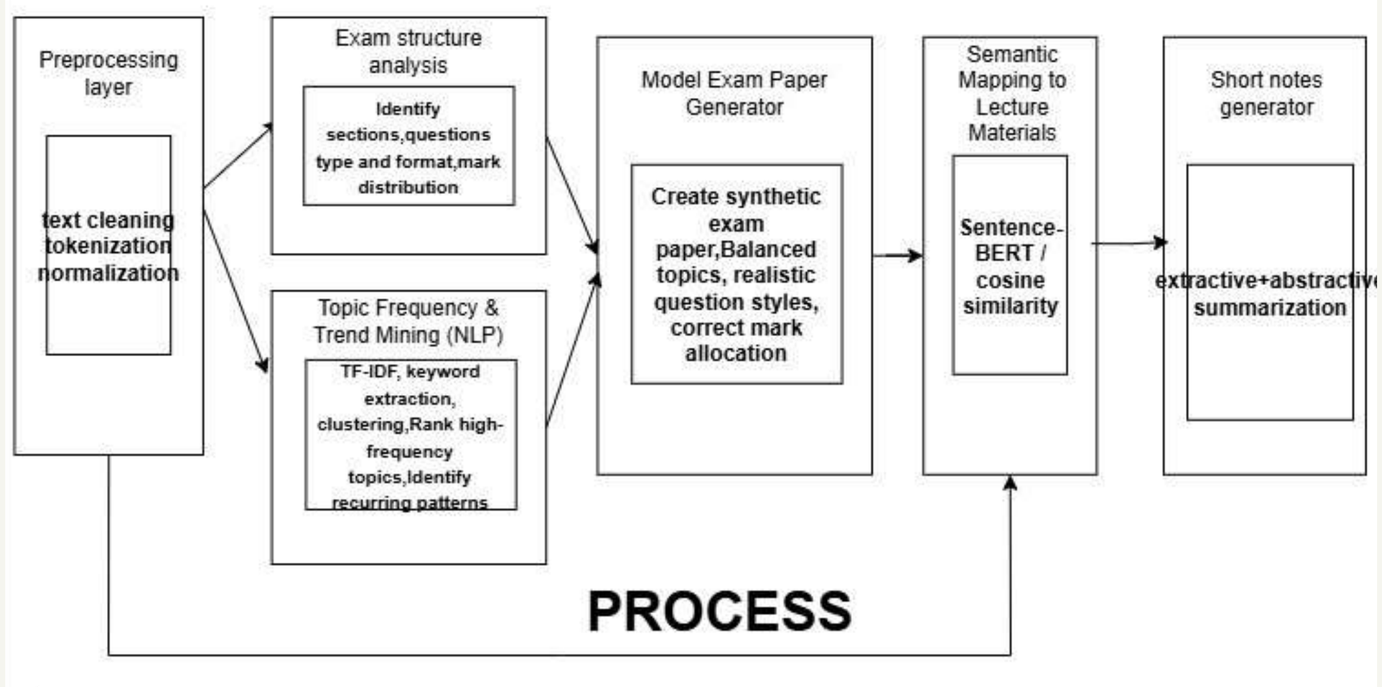
Short Note Generation
- TextRank (pyTextRank) → extractive summarization
- BART / PEGASUS → abstractive summarization

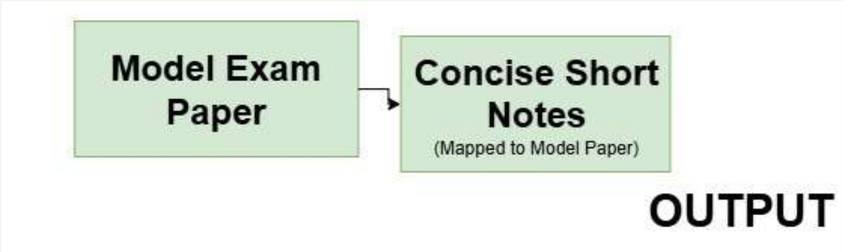
Data Availability & Ethical Clearance

- Lecture slides & past papers of DMS module(institution-provided).
- All data is academic, non-personal, no privacy risks.
- Ethical clearance obtained for exam papers & lecture content usage.

SYSTEM OVERVIEW







REAL WORLD SCENARIO

Student who plans to study for the exam

Attempt the Model Exam Paper with the help of the blueprint

Struggle to find the answer?
or
need to learn about that concept

Open concise notes for that topic

Flag the question

REFERENCE

[5] A. Kurdi, J. Leo, B. Parsia, U. Sattler, and S. Al-Emari, "A systematic review of automatic question generation for educational purposes," International Journal of Artificial Intelligence in Education, vol. 30, no. 1, pp. 121-204, 2020. [Online]. Available: https://www.researchgate.net/publication/337433098_A_Systematic_Review_of_Automatic_Question_Generation_fo r_Educational_Purposes

[6] H. Kurdi, R. El-Sayed, and A. A. El-Khatib, "Automatic question generation and answer assessment: A survey in technology enhanced learning," Smart Learning Environments, vol. 8, no. 1, pp. 1-22, 2021. [Online]. Available: https://telrp.springeropen.com/articles/10.1186/s41039-021-00151-1

[7] Z. Duan, C. Li, X. Sun, and H. Ji, "Controllable question generation via answer-aware prompts and difficulty calibration," arXiv preprint, arXiv:2106.15313, 2021. [Online]. Available: https://arxiv.org/abs/2106.15313

[8] S. Reddy, V. Gupta, and P. Singh, "Large Language Models for educational content generation and evaluation: Opportunities and challenges," arXiv preprint, arXiv:2410.09063, 2024. [Online]. Available: https://arxiv.org/abs/2410.09063

An NLP-Driven System for Automated Study Material Generation and Continuous Assessment Guidance

Specialization - Data Science

IT22171306 | Janushiya R | 25-26J-089

RESEARCH GAP & PROVEN GAP

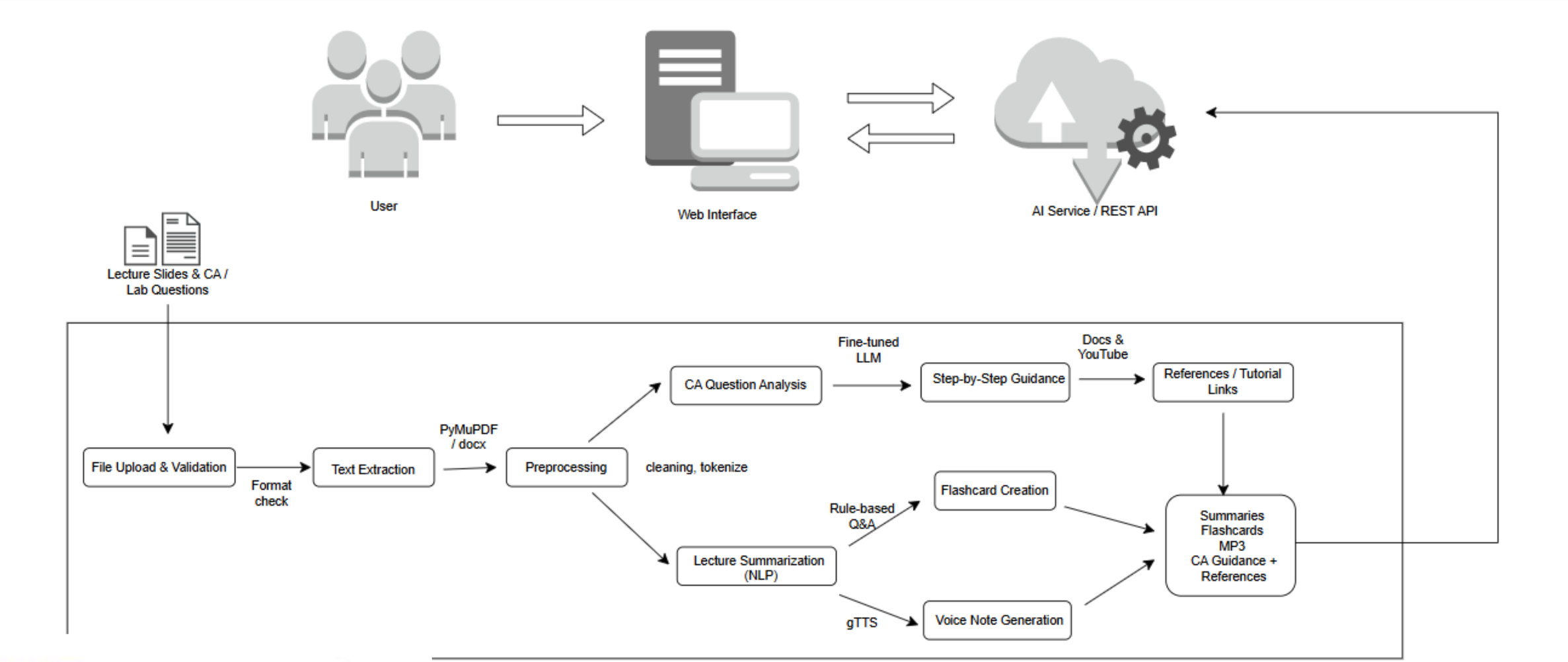
Feature	Research 01 [1]	Research 02 [2]	Research 03 [3]	Proposed System
Lecture summarization	✗	✓	✓	✓
Generate flashcards	✗	✓	✓	✓
Multimodal delivery (voice notes, flashcards)	✗	✗	✓	✓
Syllabus-aware CA guidance	✗	✗	✗	✓
Institution-specific dataset adaptation	✗	✗	✗	✓

CREATIVE SOLUTION

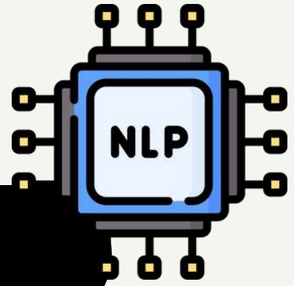
AI-powered system with:

- NLP (T5, BART) → Summarization
- Rule-based logic → Flashcards
- TTS (gTTS) → Voice notes
- Fine-tuned LLM (Gemma 3B, 1.5B) → Step-by-step CA guidance & Reference links

SYSTEM DIAGRAM



DOMAINS OF KNOWLEDGE



Natural Language Processing
(Summarization, Flashcards)

Information Retrieval
(Question bank, search)

Educational Technology
(Study support, CA guidance)

Machine Learning
(RAG, fine-tuning)

TOOLS & TECHNOLOGY

Technologies Used

NLP models - T5, BART
File Parsing Libraries - PyMuPDF, python-docx

Text-to-Speech - gTTS Library

Database - MongoDB

Other Tools - Draw.io, Git

Rule-based Logic - spaCy, Regex, and NLTK

Fine-tuned LLM - GPT, Gemma 3B, 1.5B

Front-end Framework - Streamlit

Project Management - Jira

FRAMEWORK & VALIDATION

Evaluation / Success Measures

- Summarization → ROUGE, BLEU, BERTScore + validation by subject experts
- Flashcards → Coverage of key concepts & expert review for accuracy
- Voice Notes → Speech clarity, usability feedback from students
- CA Guidance → Step-by-step correctness & user study with DMS students

Data Availability

- Lecture slides & notes from the DMS module
- Historical Continuous Assessment (CA) questions & lab tests
- Fine-tuning dataset of past university CA solutions

Ethical Clearance

- Only publicly available/authorized academic materials (no confidential data)
- Expert validation to ensure academic relevance and institutional compliance

REFERENCE

[1]: Zhang, Y., Wang, X., Zhang, Y., Li, Y. and Zhang, Y., 2025. The Power of Voice Narration for Study Notes. Scholarly Blog. Available at: https://scholarly.so/blog/the-power-of-voice-narration-for-study-notes [Accessed 12 Aug 2025].

[2]: Chen, J., Zhang, W., Liu, Y., Sun, J. and Zhao, X., 2023. KG-RAG Tutor: Knowledge Graph Enhanced Retrieval-Augmented Generation for Education. arXiv preprint arXiv:2311.17696. Available at: Transformer-based NLP models (T5, BART) .

[3]: Banerjee, A., Singh, R., Kumar, M. and Dutta, A., 2023. AI-Enabled Intelligent Assistant for Higher Education: Generating Quizzes, Flashcards, and Personalized Guidance. arXiv preprint arXiv:2309.10892. Available at: Transformer-based NLP models (T5, BART) .

An Intelligent Essay-based Learning Support System with Difficulty Prediction and Feedback-driven Improvement

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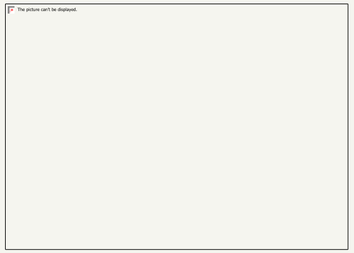
INTRODUCTION

Knowledge Gap

- Students face inefficient exam preparation due to a lack of structured practice and receive generic feedback that fails to address specific learning gaps.



Proposed Solution

- Uses NLP and transformer models to predict essay question difficulty based on lecture materials and exam data.
- Organizes past exam questions by difficulty to help students practice progressively according to their skill level.
- Evaluates essays with NLP-based scoring, providing personalized feedback and actionable suggestions for improvement.

DOMAINS OF KNOWLEDGE

- The system integrates Educational Technology to create adaptive, personalized learning platforms, while AI automates essay evaluations, predicts difficulty, and offers personalized feedback.

- NLP enables automated scoring, while Cognitive Science & Bloom's Taxonomystructure cognitive complexity in evaluations, and Educational Analytics optimizes curriculum and learning outcomes by analyzing student performance data.

TOOLS & TECHNOLOGIES

ML model

Transformer model (BERT)
XGBoost

Database

Mongo DB

Programming Languages

Python
Javascript

Project Management

Jira

Other tools

Git
Figma
Postman

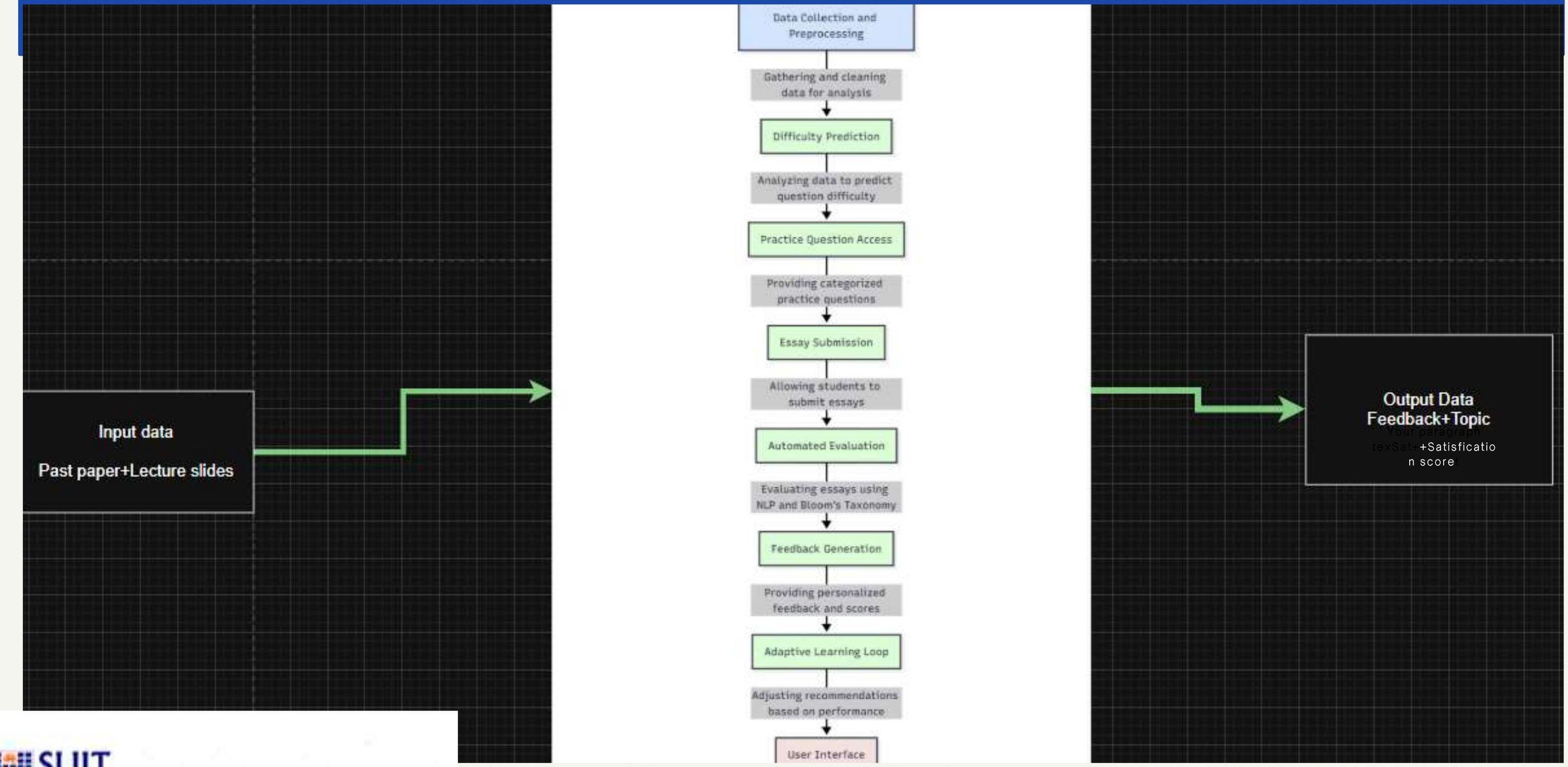
Frameworks & Libraries

Flask
React.js&
Streamlit
Tensorflow
spaCy
Pandas, NumPy
Fuzzy logic
Bloom's Taxonomy

EVALUTION/SUCCESS MEASURES

- Metrics:Prediction Accuracy, Automated Essay Scoring,Personalized Feedback Quality.
- Comparison:Rule-Based vs ML/DeepLearning vs Hybrid.
- Validation:5-fold cross-validation,Expert Evaluation, User Testing.
- Target Accuracy:> 80% for predicting essay question difficulty and automated essay scoring.

SYSTEM OVERVIEW



REAL WORLD USAGE

- Student Access via Multiple Platforms.
- Practice with Categorized Past Exam Questions.
- Essay Submission and Real-Time Feedback.

An NLP-Driven System for Semantic Classification and Personalized Study Plan Generation

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IT22113672 | Kajamalan S | 25-26J-089

INTRODUCTION

Knowledge Gap

- DMS students at SLIIT struggle to match messy lecture notes with syllabus.
- They find it difficult to link notes with exam-relevant topics
- Miss recurring exam patterns, leading to wasted effort on low priority topics

Proposed Solution

- Current AI tools focus on general learning, not exams, and lack questions tagging and personal plans.
- My solution uses a custom-Trained NLP model notes to syllabus modules, analyzes topic importance with TF-IDF, Create personalized plans with recommended practice questions, making study time more effective.

DOMAINS OF KNOWLEDGE

- NLP for semantic classification of notes and questions
- TF-IDF for frequency analysis to identify high-priority topics.
- Machine learning for analyzing exam trends and recommending questions

TOOLS & TECHNOLOGIES

Project Management

Jira

Database

Faiss
Mongo DB

Programming Languages

Python
Javascript

ML

BERT,T5/ XG Boost

Other tools

Git
Figma
Postman
Dask
Google Colab

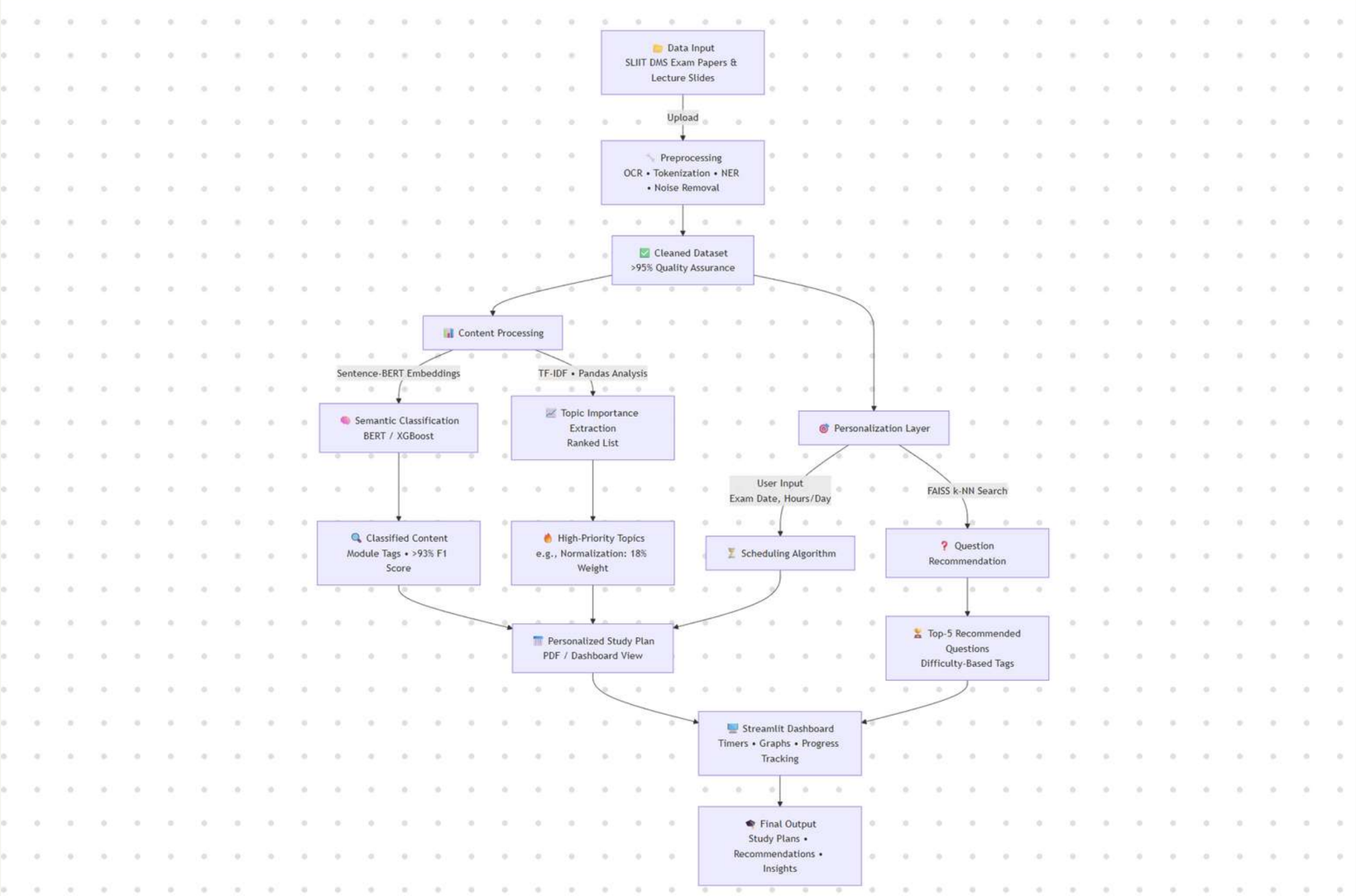
Frameworks & Libraries

Flask
React & Streamlit
Tensorflow
NLTK / spaCy
Matplotlib ,
Seaborn
Pandas, NumPy

EVALUTION/SUCCESS MEASURES

- Metrics: Precision, Recall, F1-score
- Comparison: Rule-based vs ML/Deep Learning
- Validation: 5- fold cross-validation to ensure robust performance
- Target Accuracy: > 80 %

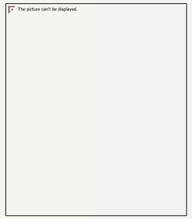
SYSTEM OVERVIEW



REAL WORLD USAGE

- Student access via phone or computer
- Input exam date and available study hours
- Receive a customized study plan with key topics (e.g “Day 1: 3h Normalization”)
- Practice with system-suggested questions
- Track progress using timers for improved exam preparation for DMS module

COMMERCIALIZATION



Advertising

Subscription

BUDGET

Type	Cost
Internet use and web hosting	LKR.10,000.00
Training Cost	LKR.10,000.00
Publication Cost	LKR.50,000.00
Stationery	LKR.1,000.00
Additional	LKR.10,000.00
Total	LKR.81,000.00

- Recovering the cost through commercialization

MARKET & VALUE PROPOSITION

- Target Market: University students
- Customer Profile: SLIIT University IT students - 2nd year
- Value Proposition:
  - Bridges curriculum delivery and exam assessments
  - Offers multiple outputs: model papers, notes, flashcards, feedback, study plans, and CA guidance
- Intellectual Property Rights:
  - Copyright: Protection for generated platform
  - Data rights: Institutional agreements for use of past papers & lecture materials.

THANK YOU